

Customer Shopping Behavior Analysis - Detailed Project Report

Executive Summary

The Customer Shopping Behavior Analysis project was developed to analyze customer purchasing trends, spending behavior, product preferences, subscription patterns, and customer satisfaction. The project utilized Python, Pandas, NumPy, Jupyter Notebook, and Power BI to transform raw customer transaction data into actionable business insights. The dataset consists of 3,900 customer records containing demographic information, purchasing behavior, review ratings, payment methods, shipping preferences, and subscription details.

Dataset Overview

Total Records: 3,900 Customers

Attributes Analyzed:

- Age
- Gender
- Product Category
- Purchase Amount
- Subscription Status
- Shipping Type
- Payment Method
- Purchase Frequency
- Review Ratings
- Discount Usage
- Promotional Code Usage
- Previous Purchases

Key Findings

1. Male customers represent approximately 68% of the customer base.
2. Clothing is the most popular and highest revenue-generating category.
3. Total Revenue generated is \$233,081.
4. Average Purchase Amount is approximately \$59.76.
5. Clothing and Accessories contribute around 77% of total revenue.
6. Only 27% of customers are subscribed to the membership program.
7. Sales remain relatively stable throughout all seasons.
8. Customer review ratings indicate overall positive customer satisfaction.

Revenue Analysis

Revenue by Category:

- Clothing: \$104,264
- Accessories: \$74,200
- Footwear: \$36,093
- Outerwear: \$18,524

Observation:

Clothing is the strongest-performing category while Outerwear contributes the least revenue and requires improvement strategies.

Business Insights

- Clothing and Accessories are the primary revenue drivers.
- Subscription adoption remains low, creating an opportunity for customer retention programs.
- Female customer participation can be expanded through targeted marketing.
- Outerwear sales require strategic improvements.
- Stable seasonal demand allows easier inventory forecasting and planning.

Recommendations to the Company

1. Strengthen Subscription Programs
 - Offer exclusive discounts.
 - Introduce reward points.
 - Provide early access to new products.
2. Expand High-Performing Categories
 - Increase inventory in Clothing and Accessories.
 - Launch premium product collections.
 - Invest in category-specific marketing.
3. Improve Outerwear Sales
 - Bundle products with Clothing.
 - Offer seasonal promotions.
 - Improve product visibility.
4. Personalized Marketing
 - Age-based recommendations.
 - Gender-specific campaigns.
 - Purchase-history-based promotions.

5. Leverage Customer Reviews

- Promote highly-rated products.
- Address customer complaints quickly.
- Continuously collect feedback.

6. Implement Loyalty Programs

- Cashback rewards.
- VIP memberships.
- Referral benefits.

Business Impact

The implementation of these recommendations can result in:

- Increased customer retention.
- Higher subscription adoption rates.
- Improved customer satisfaction.
- Increased sales and profitability.
- Better inventory management.
- Data-driven decision making.

Tools & Technologies Used

- Python
- Pandas
- NumPy
- Jupyter Notebook
- Power BI
- Data Visualization Techniques

Conclusion

The Customer Shopping Behavior Analysis project successfully identified customer trends, purchasing patterns, revenue drivers, and growth opportunities. The findings suggest that focusing on customer retention, subscription growth, personalized marketing, and optimization of high-performing categories can significantly improve business performance and customer experience.